

COURSE TITLE : **MICROPROCESSORS AND INTERFACING**
COURSE CODE : **4151**
COURSE CATEGORY : **A**
PERIODS/WEEK : **4**
PERIODS/SEMESTER : **52**
CREDITS : **4**

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Basic 80x86 Architecture	13
2	Programming of x86 processor	13
3	Interrupt mechanism of x86 & Interfacing of chips	13
4	Advanced Processor Technologies	13

Course General Outcomes:

Sl.	G.O	On completion of this course the student will be able :
1	1	To understand Microprocessors
	2	To know the architecture of 8086 Microprocessor
2	1	To understand the Programming of 8086
	2	To understand The Instruction set of x86 Processor
3	1	To understand the Interrupt mechanism of x86
	2	Interfacing with PPI & Keyboard and Display Controllers
4	1	To understand the operating Modes of 80386
	2	To study Pentium Processor and modern technologies

Specific Outcomes:

MODULE –I Basic 80x86 Architecture

1.1 To understand Microprocessors

- 1.1.1 To describe the role of Microprocessor in Micro Computer
- 1.1.2 To list the features of 8086

1.2 To know the architecture of 8086 Microprocessor

- 1.2.1 To Explain the architecture of 8086
- 1.2.2 To describe the software architecture of 8086
- 1.2.3 To describe the Hardware structure of 8086

MODULE – II Programming of x86 processor

2.1 To understand the Programming of 8086

- 2.1.1 To Describe Assembly language programming

2.2 To understand The Instruction set of x86 Processor

- 2.2.1 To describe data transfer, branch, and arithmetic instructions
- 2.2.2 To describe logical, shift , and rotate instruction
- 2.2.3 To describe string instructions and procedures

MODULE – III Interrupt mechanism of x86 & Interfacing of chips

3.1 To understand the Interrupt mechanism of x86

- 3.1.1 To describe interrupts of 8086
- 3.1.2 To explain the different types of interrupts

3.2 To Understand different Interfacing Chips

- 3.2.1 To describe Programmable Peripheral Interfacing Chip and Interfacing with x86 processor
- 3.2.2 To explain the importance, organisation, and interfacing of Programmable Interrupt Controller
- 3.2.3 To describe Keyboard and Display Interface chip and Interfacing with x86 processor

MODULE –IV Advanced Processor Technologies

4.1 To understand the operating Modes of 80386

4.0.1 To list the features of 80386

4.1.2 To describe Real Mode & Protected Virtual Addressing Mode

4.1 To Explain Pentium Processor

4.1.1 To list the features of Pentium

4.1.2 To describe pipelining

4.2 To know the advanced technologies of modern Intel processors

4.2.1 To define Super scalar Architecture

4.2.2 To define Multicore processing

4.2.3 To define MMX Technology

4.2.4 To define Hyperthreading

CONTENT DETAILS

MODULE –I Basic 80x86 Architecture

Role of Microprocessor in Micro Computer – Brief history of Microprocessors (with specific insight into x86 family) - Features of 8086

Internal Block Diagram of 8086 – Execution Unit – Bus Interface Unit – Addressing Modes

Hardware structure of 8086- Pin Configuration-Clock- Processor activities (Interrupt, DMA, etc.)-

Maximum mode- Instruction cycle

Assembly process – Assemblers for x86 – Instruction Design

MODULE – II Programming of x86 processor

Data transfer Instructions- Branch instructions- Arithmetic instructions- Shift and Rotate Instructions- String Instructions- Procedures- Macros-Number Format Conversions- ASCII operations

MODULE – III Interrupt mechanism of x86 & Interfacing of chips

Interrupts of 8086- Dedicated Interrupt types- Software interrupts-Hardware interrupts- Priority of interrupts-Programmable Interrupt Controller (8259)

Organisation and Interfacing of PPI (8255), and Keyboard and display Interface (8279)

MODULE –IV Advanced Processor technologies

Features of 80386- Real Mode - Protected Virtual Addressing Mode

Features of Pentium- pipelining- Stages of pipelining- Speedup due to pipelining- Pipeline Hazards

Super scalar Processors- Multiple Execution units

Multicore processing – Major issues in Multicore Processing (interconnects- cache coherence- snooping protocol- Directory based protocol)

MMX- SSE- Hyperthreading

Note: The programming should be covered irrespective of Assembler.

Text Book(s) :

1. The x86 Microprocessors- Architecture, Programming and Interfacing – Lyla B Das – Pearson- Second edition.
2. Microprocessor and Microcontroller - R. Theagarajan – SCITECH-2010

References

1. The 8088 and 8086 Microprocessors – Programming, Interfacing, Software and Hardware Applications by Walter A. Triebel & Avatar Singh, Pearson Fourth Edition,
2. Microprocessor 8086 Architecture, Programming and Interfacing by Sunil Mathur, PHI, 2011
3. The Intel Microprocessors : Architecture, Programming and Interfacing- Barry B. Brey Pearson -8 Edition